

AOS-CX 10.16.xxxx Containers Guide

**(6300F/M, 6400, 8xxx, 9xxx, 100xx Switch
Series)**



**Hewlett Packard
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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, S0X44A, S3L75A, S3L76A, S3L77A, S4P41A, S4P42A, S4P43A, S4P44A, S4P45A, S4P46A, S4P47A, S4P48A)
- HPE Aruba Networking 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A, S0E48A #0D1, S1T83A, S1T83A #0D1)
- HPE Aruba Networking 8100 Switch Series (R9W94A, R9W95A, R9W96A, R9W97A)
- HPE Aruba Networking 8320 Switch Series (JL479A, JL579A, JL581A)
- HPE Aruba Networking 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- HPE Aruba Networking 8325H Switch Series (S4B20A, S4B21A, S4B22A, S4B23A, S2T42A, S2T46A, S2T47A, S2T48A)
- HPE Aruba Networking 8325P Switch Series (S0G12A, S4A48A)
- HPE Aruba Networking 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A, JL700C, JL701C, JL702C, JL703C, JL706C, JL707C, JL708C, JL709C, JL710C, JL711C, JL704C, JL705C, JL719C, JL718C, JL717C, JL720C, JL722C, JL721C)
- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)
- HPE Aruba Networking 9300 Switch Series (R9A29A, R9A30A, R8Z96A, S0F81A, S0F82A, S0F83A)
- HPE Aruba Networking 9300S Switch Series (S0F81A, S0F82A, S0F83A, S0F84A, S0F85A, S0F86A, S0F88A, S0F95A, S0F96A)
- HPE Aruba Networking 10000 Switch Series (R8P13A, R8P14A)
- HPE Aruba Networking 10040 Switch Series (S4R58A, S4R59A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ▪ <code><example-text></code> ▪ <code><example-text></code> ▪ <code>example-text</code> ▪ <code>example-text</code>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ▪ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ▪ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ▪ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ▪ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where **<VLAN-ID>** is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 6300 Switch Series

- **member**: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- **slot**: Always 1. This is not a modular switch, so there are no slots.
- **port**: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the HPE Aruba Networking 6400 Switch Series

- **member**: Always 1. VSF is not supported on this switch.
- **slot**: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- **port**: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

On the HPE Aruba Networking 8xxx, 93xx, and 10xxx Switch Series

- **member**: Always 1. VSF is not supported on this switch.
- **slot**: Always 1. This is not a modular switch, so there are no slots.
- **port**: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.



If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Container overview

The Container Manager feature allows containers to run inside an AOS-CX switch. It allows the switch to launch multiple container images, whether HPE-signed or unsigned, and provides tools to manage, maintain, and monitor the lifecycle of these images and containers. Applications can be run within the container infrastructure, turning them into containerized applications.

The AOS-CX container infrastructure is designed to support containers that run in the background as services or scheduled jobs. These containers either keep running continuously or stop gracefully after completing their tasks. Examples of supported workloads include web servers like Nginx or Apache, application servers built with Node.js, Java, or Python APIs, and system monitoring tools.

The container infrastructure does *not* support containers that must interact users in real time through a command-line interface, such as shells (bash, sh), interpreters (Python, Node.js), or text-based tools (like vim or top) to accept user input and display output interactively. If you try to deploy an unsupported container that relies on an interactive terminal, it may stop immediately or behave unpredictably.

Supported switches

The following switch platforms are validated for data center deployments. The agent itself has no context about the switch that it is installed on, except that the chipset architecture must be either **x86_64** or **arm**. The container image size is limited by platform. A container instance will display an error if you attempt to download a container image that exceeds the supported image size.

Switch	Chipset Architecture Type	Maximum Image size (MiB)
6300	arm	500
6400	arm	500
8100	arm	500
8320	x86_64	1000
8325	x86_64	1000
8325H	x86_64	1000
8325P	x86_64	1000
8360	arm	500
8400	x86_64	500
9300	x86_64	1000
9300S	x86_64	1000

Switch	Chipset Architecture Type	Maximum Image size (MiB)
10000	x86_64	1000
10040	x86_64	1000



The HPE Aruba Networking CX vSphere agent is a containerized agent developed for data center applications. The plugin consumes memory and CPU that would otherwise be allocated to the switch. Thus, the targeted platforms have higher system resources. Those resources are capped to prevent its interference with performance of the switch.



The solution was tested and qualified on the platforms listed here. Other platforms may work at low scale, but they are not recommended outside of the lab or test environments due to the resource requirements for vSphere.

USB persistent storage

Containers can mount additional persistent storage via a USB device using the [mount](#), [source](#) and [destination](#) commands to create a mount ID and specify the mount source and destination paths. Best practices is to configure persistent storage via these commands. If a USB directory in the source filesystem is deleted or renamed through shell commands, the container may not function correctly.

The USB persistent storage feature includes the following additional caveats.

- Containers are not aware if the configured USB device is unmounted, disabled or physically removed. As a result, the container's correct functionality can't be guaranteed if these actions are made without unconfiguring the container mount first.
- For containers using USB persistent storage functionality, data won't synchronize between the management modules after a High Availability event, so the container deployed on the new management module won't have access to the USB device connected to the previous management module.
- If there is an issue with the USB device mount when the container boots up, there can be a potential wait of up to two minutes before the container shows an error.

Restart protection

The HPE Aruba Networking switch can prevent a crashing containerized application from restarting after three restarts within a brief period of time. Once the restart protection feature indicates that it has stopped a crashing application in this manner, the container will display a **run failed** status in the output of the **show container** command.

If the configured image has been downloaded and verified by container manager, in case of a restart, the image will not need to be downloaded again unless the image location is modified. This also applies when using the **allow-unsigned** parameter, if a re-download is required, modify the image location URL or remove the command and add it again.

vSphere agent

The HPE Aruba Networking CX vSphere agent allows administrators to trace how Virtual Machines are connected to the physical switches. The ability to visualize how a Virtual Machine connects to a CX port helps to resolve connectivity issues. This functionality helps operators in several troubleshooting scenarios, such as diagnosing a connectivity issue, or ensuring adequate distribution of VMs across the physical infrastructure.

You can install the vSphere agent on one or many of the switches in the fabric. You can perform the following tasks, from any switch where the agent is installed:

1. Run the `show VMs` command to list all the VMs managed by the vCenter Appliance integrated with the agent. This functionality also helps to view VMs connections to the ports on other switches on the fabric.
2. Search for a specific VM by name. To find the connection between the VM and a switch port, attach the switch to at least one host managed by the integrated vCenter.



The **show neighbors** command examines the LLDP data that is advertised from any switch to a distributed virtual switch and provides connection details for third-party switches.

For technical information on containers, including use cases, example configurations and best practices, view the [AOS-CX Container Enhancements video on the HPE Aruba Networking broadcasting channel on YouTube](#). For more information about cloud capabilities of vSphere agent, refer to [vSphere Integration](#).

Installation and Configuration of the vSphere agent

To install a container, perform the following steps:

1. Download the vSphere agent from the HPE Aruba Networking support portal and select the **arm** or **x86_64** architecture for the switch platform. Refer to the list of [supported switches](#) to select the corresponding architecture.
2. Host the agent on a local network that can be accessed using the HTTP protocol. Limitations of the container framework prevent usage of the HTTPS protocol. The HTTP server must be reachable on the mgmt or any other configured VRF.

To create the container configuration:

1. Enter the **config** command to enter the config context.
2. Create and enter the **config-container-<name>** context using the [container](#) command.
3. From the **config-container-<name>** context, configure the container image location using the [image-location](#) command. To create a container configuration that bypasses signature validation, include the optional **allow-unsigned** parameter for the image-location command
4. (Optional) Configure a container environment variable using the [env](#) command.

5. (Optional) To create a configuration with VRF network connectivity, issue the [network vrf](#) command, then configure port mapping using the [port-map](#) command.
6. (Optional) To create a configuration that limits CPU usage to a percentage of the total switch CPU, use the [restrict cpu](#) command.
7. (Optional) To create a configuration that limits memory resources to maximum number of megabytes (up to 20% of the total switch memory), use the [restrict memory](#) command.
8. (Optional) To create a configuration with mounted storage, issue the [mount](#) command, then use the [source](#) and [destination](#) parameters from within the **config-container-mount-<id>** context to define the mount source and destination.
9. From the **config-container-<name>** context, enable the container using the [enable](#) command.
10. Issue the [show container](#) command to validate the container status.

Show commands

There are four top-level show commands.

- vms - shows virtual machine connection information.
- connection-status - shows information about configured vSphere vCenters.
- neighbors - shows learned neighbor information (Note: This information is learned from LLDP on vCenter distributed vswitches, not the switch).
- help - shows agent version, description, and configurable environment variables.

show vms

The following show command describes the status of the connection to vSphere, and show the VMs that are attached to switches:

At least one command parameter must be used if the parameters are available.

For example, with the show vms command, we force use of the brief option so the user is aware of all available options.

If not specified, all commands default to brief.

```
igor-sw-02# container vsphere exec show vms brief Hypervisor VM Name Physical
Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 vm2 laser-
sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 vm5 laser-sw-02 (b8:d4:e7:da:40:00) 1/1/4
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/3 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3
laser-01-esxi.lab.plexxi.com vm1 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/1 vm3 laser-
sw-01 (b8:d4:e7:d9:af:00) 1/1/2
laser-sw-02 (b8:d4:e7:da:40:00) 1/1/1
igor-sw-02# container vsphere exec show vms vm-name vm4 Hypervisor VM Name
Physical Switch Connections Interface

laser-02-esxi.lab.plexxi.com vm4 laser-sw-01 (b8:d4:e7:d9:af:00) 1/1/3 igor-sw-02#
container vsphere exec show vms vm-name vm4 detailed VM Name : vm4
Hypervisor : laser-02-esxi.lab.plexxi.com Vendor : vmware
Hostname : vm4
OS : CentOS 4/5 or later (64-bit) VM status : on
Virtual NIC : Network adapter 2 IP Address : 172.20.12.204
MAC Address : 00:50:56:89:3c:bf Virtual Switch : dvs4
Port Group : dp4 Vlan : 0
```

```
Physical Switch Connections Interface : 1/1/3
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : HPE_ANW JL635A GL.10.12.0001G Management Address :
172.20.11.253

Container exec commands are top-level, and can be run in any CLI context.

igor-sw-02# container vsphere exec show connection-status brief Hypervisor Last
Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:03 connected igor-sw-02# config
igor-sw-02(config)# container vsphere exec show connection-status brief Hypervisor
Last Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:07 connected igor-sw-02(config)#
interface 1/1/1
igor-sw-02(config-if)# container vsphere exec show connection-status brief
Hypervisor Last Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-17 23:30:15 connected
```

show vms

This command displays the connectivity status for the vSphere agent..

The connection timestamp is updated whenever a successful connection is made with the vSphere agent. The sync timestamp is updated whenever a full sync, event poll, or event process occurs.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status brief Hypervisor Last Sync Connection State

laser-vcsa.lab.plexxi.com 2023-03-18 02:36:06 connected
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : connected Last Sync : 2023-03-18 02:35:08
Last Connection : 2023-03-18 02:35:09
igor-sw-02(config-container-vsphere)# container vsphere exec show connection-
status detailed Hypervisor : laser-vcsa.lab.plexxi.com
Connection State : disconnected Last Sync : 2023-03-18 02:04:05
Last Connection : 2023-03-18 02:09:30
Connection Failure Reason : Could Not Resolve Hostname
```

show neighbors

show neighbors command displays the virtual network adapter neighbor information.

Neighbors are pulled from distributed virtual switches of the vSphere and cached locally. Neighbors are used to map VMs to physical switch ports.

If you do not see a neighbor on the port, you will not see any VM connectivity on that port either. Detailed and brief options are available. Filtering can be done on switch-name and/or interface.

```
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors brief
Chassis ID Switch Name Interface

b8:d4:e7:da:40:00 laser-sw-02 1/1/1 1/1/2
b8:d4:e7:d9:af:00 laser-sw-01 1/1/1 1/1/2
igor-sw-02(config-container-vsphere)# container vsphere exec show neighbors
detailed Chassis ID : b8:d4:e7:da:40:00
```

```

Switch Name : laser-sw-02
Switch Description : HPE_ANW JL635A GL.10.10.1000 Management Address :
192.168.200.1
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102
Chassis ID : b8:d4:e7:d9:af:00 Switch Name : laser-sw-01
Switch Description : HPE_ANW JL635A GL.10.10.1000 Management Address :
192.168.200.0
Interface : 1/1/1 Advertisement Type : lldp TTL : 102
Interface : 1/1/2 Advertisement Type : lldp TTL : 102

```

Supported switches

The following switch platforms are validated for data center deployments. The agent itself has no context about the switch that it is installed on, except that the chipset architecture must be either **x86_64** or **arm**. The container image size is limited by platform. A container instance will display an error if you attempt to download a container image that exceeds the supported image size.

Switch	Chipset Architecture Type	Maximum Image size (MiB)
6300	arm	500
6400	arm	500
8100	arm	500
8320	x86_64	1000
8325	x86_64	1000
8325H	x86_64	1000
8325P	x86_64	1000
8360	arm	500
8400	x86_64	500
9300	x86_64	1000
9300S	x86_64	1000
10000	x86_64	1000
10040	x86_64	1000



The HPE Aruba Networking CX vSphere agent is a containerized agent developed for data center applications. The plugin consumes memory and CPU that would otherwise be allocated to the switch. Thus, the targeted platforms have higher system resources. Those resources are capped to prevent its interference with performance of the switch.



The solution was tested and qualified on the platforms listed here. Other platforms may work at low scale, but they are not recommended outside of the lab or test environments due to the resource requirements for vSphere.

USB persistent storage

Containers can mount additional persistent storage via a USB device using the [mount](#), [source](#) and [destination](#) commands to create a mount ID and specify the mount source and destination paths. Best practices is to configure persistent storage via these commands. If a USB directory in the source filesystem is deleted or renamed through shell commands, the container may not function correctly.

The USB persistent storage feature includes the following additional caveats.

- Containers are not aware if the configured USB device is unmounted, disabled or physically removed. As a result, the container's correct functionality can't be guaranteed if these actions are made without unconfiguring the container mount first.
- For containers using USB persistent storage functionality, data won't synchronize between the management modules after a High Availability event, so the container deployed on the new management module won't have access to the USB device connected to the previous management module.
- If there is an issue with the USB device mount when the container boots up, there can be a potential wait of up to two minutes before the container shows an error.

Restart protection

The HPE Aruba Networking switch can prevent a crashing containerized application from restarting after three restarts within a brief period of time. Once the restart protection feature indicates that it has stopped a crashing application in this manner, the container will display a **run failed** status in the output of the **show container** command.

If the configured image has been downloaded and verified by container manager, in case of a restart, the image will not need to be downloaded again unless the image location is modified. This also applies when using the **allow-unsigned** parameter, if a re-download is required, modify the image location URL or remove the command and add it again.

Container management commands

container

```
container <CONTAINER-NAME>
no container <CONTAINER-NAME>
```

Description

Enters into the container configuration context.

The **no** form of this command removes the existing configurations of the specified container.

Parameter	Description
<code><CONTAINER-NAME></code>	The container name can contain up to 64 characters. Only letters, numbers and underscore (_) characters are permitted

Example

Configures a new container:

```
switch(config)# container app
The feature being used requires a AOS-CX Advanced Software Feature Pack.
For more information, refer to the AOS-CX Feature Pack Deployment Guide.
```

AOS-CX does not enforce the requirement to own a feature pack prior to using container features. This warning message is displayed only during creation, subsequent calls to the container context will not display the message.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	config config-container- <code><CONTAINER-NAME></code>	Administrators or local user group members with execution rights for this command.

container exec

```
container <NAME> exec <PARAMS>
```

Description

Allows the execution of an endpoint script in the container. The location of this endpoint is provided to the container manager infrastructure through a manifest file in the image file system of the container. This manifest file provides metadata related to the container application. When the exec command

runs, the manifest information is used to determine the endpoint to execute and the user parameters are passed directly to the endpoint. The output of such execution is provided directly to the user through the CLI. In case the manifest information or the endpoint file are missing an error is presented to the user. The user can interrupt the execution by pressing **Ctrl+C**.



If the container is not operational when the command is executed, the following error message is returned:

Failed to execute endpoint - The container is not operational.

Parameter	Description
<NAME>	Specifies a container name up to 64 characters long.
exec	Runs a container application command.
<PARAMS>	Specifies container command parameters.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	config-container-<CONTAINER-NAME>	Administrators or local user group members with execution rights for this command.

enable (container manager)

enable
no enable

Description

A container is disabled by default when it is created. Use the **enable** command to enable the container.

The **no** version of this command disables the container. When an enabled container is disabled, system resources are released but the container configuration settings are stored.

Example

Enabling and then disabling the container **test**:

```
switch(config-container-test)# enable
switch(config-container-test)# no enable
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	config-container- <i><CONTAINER-NAME></i>	Administrators or local user group members with execution rights for this command.

env

env *<NAME>* {value *<VALUE>*}[encrypted [plaintext *<VALUE>*|ciphertext *<VALUE>*]]
no env *<NAME>* {value *<VALUE>*}[encrypted {plaintext|ciphertext}*<VALUE>*]

Description

Configures an environment variable for a container that is composed of a key and a value pair. The key-value pair defines the behavior of the environment in a container and is used by the container processes. The value of the environment variable can be stored in the host system as an encrypted value. The container manager infrastructure provides the decrypted value to the container. The **no** form of this command removes the configured environment variable from a container.



Configuring the **env** variable for an already operational container causes the container to restart if the container has already been enabled.

Parameter	Description
<code><NAME></code>	Specifies the name of the container environment variables.
<code>value <VALUE></code>	Specifies the variable value.
<code>encrypted</code>	Encrypts the environment variable value. If you press <enter> after the encrypted parameter, you will enter a variable configuration mode that allows you to securely enter a hidden value. <i>This is the recommended method for entering an encrypted variable</i>
<code>plaintext <VALUE></code>	Optional. Specifies the variable value in plain text. <i>Not recommended for encrypted variables.</i>
<code>ciphertext <VALUE></code>	Optional. Specifies the variable value as previously encrypted text. <i>Recommended for encrypted variables; specify the encrypted variable value as previously encrypted text.</i>

Example

Securely entering an encrypted variable:

```
6300(config-container-test)# env TEST encrypted
Enter environment variable value: *****
6300(config-container-test)# end
```

Command History

Release	Modification
10.13.1000	The plaintext and ciphertext options for the encrypted parameter are now optional. Starting with this release, you can use the encrypted option to encrypt the environment variable and specify the value in plaintext hidden from the CLI.
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360	<code>config-container-<CONTAINER-NAME></code>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
8400 9300 9300S 10000 10040		

image-location

```
image-location <URL> vrf <VRF-NAME> [allow-unsigned]
no image-location <URL> <VRF-NAME> [allow-unsigned]
```

Description

Configures the image location for a container. Modifying image location prompts an image upgrade. The **no** form of this command removes the configured location of a container.



- If the user sets a location value which does not follow the standard URL format, the following error message is returned: **Failure to configure image location: Invalid URL**
- If the user tries to use a VRF value that doesn't exist on the switch, the following error message is returned: **Failure to configure image location: Invalid VRF**
- If the image of the container exceeds the maximum image size supported by the switch, the container won't be deployed.

By default, only container images with a valid HPE signature are allowed. To bypass this signature check and allow unsigned container images, include the **allow-unsigned** parameter when you define the image location. The **allow-unsigned** parameter cannot be used if you have issued the **secure-mode enhanced** command to set the switch to enhanced secure mode.

Parameter	Description
image-location	Configures the URL of the container application.
URL	Specifies the URL of the container application. URL supports HTTP protocol. The image-location URL can either be IPv4 or IPv6 address. The IPv6 address must be provided within square brackets.
vrf <VRF-NAME>	(Optional) Specifies the VRF of the image URL.
allow-unsigned	(Optional) Allow download and deployment of an unsigned container image.

Examples

Configures the image location for the IPv4 setting:

```
switch(config-container-1)# image-location http://10.0.0.1/container.img vrf mgmt
```

Appends the port to the address if the image server is running on a port other than HTTP for an IPv4 setting:

```
switch(config-container-1)# image-location http://10.0.0.1:9050/container.img vrf mgmt
```

Configures image location for IPv6 setting by wrapping IP address between square brackets:

```
switch(config-container-1)# image-location http://[2001::2]/container.img vrf mgmt
```

Specifies port number by appending it with the IPv6 address:

```
switch(config-container-1)# image-location http://[2001::2]:9050/container.img vrf mgmt
```

When you include the **allow-unsigned** parameter on a switch in standard secure mode, the following message will be displayed to inform this can be a potential security issue.

```
switch(config-container-1)# image-location http://10.0.0.1/container.img vrf mgmt
allow-unsigned
Allowing unsigned container images poses a potential security risk
that can impact both the current device and the entire network. By
allowing installation of unsigned applications you are acknowledging
and accepting these risks. HPE shall not be responsible for the
consequences of your actions and disclaims any and all liability.

Continue (y/n)? y
```

When you attempt to include the **allow-unsigned** parameter on a switch in enhanced secure mode, the following message will appear to indicate that this parameter is not supported.

```
switch(config-container-1)# image-location http://10.0.0.1/container.img vrf mgmt
allow-unsigned
Unsigned images are not permitted in the current secure mode, using the
allow-unsigned parameter will have no effect.
```

Release	Modification
10.14	The allow-unsigned parameter is introduced.
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325	config-container- <i><CONTAINER-NAME></i>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
8325H 8325P 8360 8400 9300 9300S 10000 10040		

mount (container manager)

```
mount <id>
    destination <path>
    no ...
    source <TYPE> <PATH>
```

Description

Mount persistent storage for a container application, and optionally indicate the mounted storage source and destination paths.

Parameter	Description
<ID>	ID assigned to a mounted storage instance for a container application. Range: <1-4294967295>
destination <PATH>	The destination path is a directory or file inside the container file system. The path must use the syntax / {FILE DIRECTORY} . Only letters, numbers, slashes, underscores, dashes, and periods are allowed.
no ...	Negates any configured parameter or removes a source or destination path.
source	Define the mounted storage source.
<type>	This version of AOS-CX supports only a usb storage type. NOTE: For containers using USB persistent storage functionality, ext4 is the recommended formatting method for USB storage devices. While FAT32 is currently supported, it may be deprecated in future releases.
<PATH>	The mounted storage source path must be a valid file or directory on the USB device and is relative to the root directory of the device file system. The path must use the syntax / {FILE DIRECTORY} . Only letters, numbers, slashes, underscores, dashes, and periods are allowed.

Example

Creating the mount ID 1 for the container **test**.

```
switch(config-container-test)# mount 1
```

Configuring container mount **1** with a **USB** source and the path **/app1-data**.

```
switch(config-container-mount-1)# source usb /app1-data
```

Configuring container mount **1** with the destination path **/app1-data**.

```
switch(config-container-mount-1)# destination /app1-data
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
8100 8320 8325 8325H 8325P 8360 9300 9300S 10000 10040	config-container- <i><CONTAINER-NAME></i>	Administrators or local user group members with execution rights for this command.

network vrf

```
network vrf <VRF>  
no network vrf <VRF>
```

Description

Use this command to associate a VRF with the container and allow connectivity through the VRF. When configured, the VRF will be accessible to the container through a network namespace.

Usage

Only one network VRF can be configured per container. If an additional VRF is configured, the container deployment will fail and generate an error message indicating that the container has more than one network VRF. Modifying a network container VRF will modify the container network configuration and the container will be restarted to apply the new changes if the container has already been enabled.

Examples

The following example defines the VRF **mgmt** for the **app** container.

```
switch(config-container-app)# network vrf mgmt
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	<code>config-container-<CONTAINER-NAME></code>	Administrators or local user group members with execution rights for this command.

port-map (container manager)

`port-map host-port <PORT> container-port <PORT> protocol (tcp | udp)`

Description

Issue this command from the **config-container-network-vrf-<name>** context to define a port map from a port on the host switch to a port on the container. This will allow communication from external devices to services provided by the container.

Parameter	Description
<code>host-port <port></code>	The host switch port number. Range: 1-65535
<code>container-port <port></code>	The container port number. Range: 1-65535
<code>tcp</code>	Communications will use the TCP protocol.
<code>udp</code>	Communications will use the UDP protocol.

Usage

A port map is an optional configuration for a container network VRF, but a port mapping is required to expose the mapped port to the outside network. This command can be issued multiple times to define more than one port map for the container.



Host port conflicts are not allowed. The same host port cannot be used for different containers in the same VRF, and the host port used for the VRF cannot be reserved for a non-container application on the host switch.

Examples

Configure a port map from the host to the container for TCP traffic, where traffic to TCP port 8000 on the switch will be directed to TCP port 9000 in the container.

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000 container-port 9000 protocol tcp
```

The following example maps different ports on the host to the same container port for TCP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000 container-port 9000 protocol tcp
switch(config-container-network-vrf-mgmt)# port-map host-port 8001 container-port 9000 protocol tcp
```

The following example maps the same port on the host switch to the same container port for TCP and UDP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000 container-port 9000 protocol tcp
switch(config-container-network-vrf-mgmt)# port-map host-port 8000 container-port 9000 protocol udp
```

The following example maps the same port on different VRFs on the host switch to the same container port for TCP traffic:

```
switch(config-container-network-vrf-mgmt)# port-map host-port 8000 container-port 9000 protocol tcp
switch(config-container-network-vrf-default)# port-map host-port 8000 container-port 9000 protocol tcp
```

When removing a port mapping, if the command parameters do not match a configured port mapping, the command output displays the following message:

```
switch(config-container-network-vrf-mgmt)# no port-map host-port 8001 container-port 9000 protocol tcp
The specified port mapping is not configured
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400	config-container- <i><CONTAINER-NAME></i>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040		

restrict cpu

```
restrict cpu <PERCENTAGE>
no restrict cpu
```

Description

Configures limitations for the container CPU usage. The CPU constraint is set as a percentage of the total switch CPUs. One container or any combination of containers configured on a switch can use up to 20% of the total CPU capacity of the device.



Configuring the CPU constraint for an already operational container will cause the container to restart if the container has already been enabled.

The **no** form of this command removes restrictions on the CPU usage, and reverts back to the default limitation of 10%.

Parameter	Description
<PERCENTAGE>	Specifies percentage for the container CPU usage, The default value is 10%.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320	config-container-<CONTAINER-NAME>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
8325 8325H 8325P 8360 8400 9300 9300S 10000 10040		

restrict memory

```
restrict memory <MB>
no restrict memory
```

Description

Configures limitations for memory usage of the container. The memory constraint is set in MB, and the maximum 20% of the capacity of the device can be configured for any combination of containers running on the switch. Configuring the memory constraint for an already operational container restarts the container if the container has already been enabled.

The **no** form of this command removes restrictions on the memory usage and reverts back to the default limitation of 256 MB.

Parameter	Description
<MB>	Specifies the maximum memory usage in MB. The default value is 256 MB.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400	config-container-<CONTAINER-NAME>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
9300 9300S 10000 10040		

show container

`show container [<CONTAINER-NAME>]`

Description

Shows the configuration and status information of the containers running in the system. If the container name is not specified, displays information of all the containers. When a container name is specified, displays information specific to the container.

Parameter	Description
<code><CONTAINER-NAME></code>	Specifies the name of the container for which information need to be specified.

Usage

The **Container status** field in the output of this command can display any of the following status values:

- **initializing**: Container initialization and configuration verification is in progress.
- **waiting**: Container is waiting for configuration prerequisites to be met.
- **configuration failed**: User-specified container configuration failed.
- **network configuration failed**: Container network provisioning failed.
- **acquiring**: Container is acquiring an image.
- **acquire failed**: Container failed to acquire an image.
- **deploying**: Container is deploying.
- **operational**: Container is operational (deployed and running).
- **deploy failed**: Container failed to deploy.
- **run failed**: Container failed while running.
- **exited**: Container completed its execution and exited.

Examples

The following example shows information for a configured container with a network VRF with port mappings:

```
switch# show container
Container          : appl
Container status   : operational
Manifest status    : success
Image status       : verified
Image version      : 1.0.0
Image location VRF : mgmt
```

```

Image location URL   : http://30.0.0.2:8000/container.img
CPU limit           : 10%
Memory limit        : 512MB
Environment variables:
  PYP=/usr/bin/python3
Encrypted environment variables:
  encryptedVar1
  encryptedVar2
Network:
  VRF name   : default
  Port map   :
    8080:80/tcp
    8080:8080/udp

```

The following example shows additional error messages:

```

switch# show container
Container           : app
Container status    : configuration failed
Config failure reason : Multiple definitions of environment variable PYP
Manifest status     : error
Manifest status reason : 'exec' file not found in container
Image status        : verified
Image version       : 1.0.0
Image location VRF  : mgmt
Image location URL   : http://30.0.0.2:8000/container.img
CPU limit           : 10%
Memory limit        : 512 MB
VRFs                : mgmt
Environment variables :
  PYP=/usr/bin/python3
Encrypted environment variables:
  PYP
  encryptedVar2

```

The following example shows a configured container without signature validation:

```

switch# show container appl
Container           : appl
Container status    : operational
Manifest status     : success
Image status        : allowed without signature
Image version       : 1.0.0
Image location VRF  : mgmt
Image location URL   : http://30.0.0.2:8000/container.img
CPU limit           : 10%
Memory limit        : 512 MB
Environment variables:
  PYP=/usr/bin/python3
Encrypted environment variables:
  encryptedVar1
  encryptedVar2
Network:
  VRF name   : default
  Port map   :
    8080:80/tcp
    8080:8080/udp

```

The following example shows the command out when there are no configured containers:

```
switch# show container
No containers configured
```

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	Operator (>) or Manager (#)	

show capacities containers

```
show capacities containers
```

Description

Shows the maximum number of containerized applications that can be configured in the system.

Examples

Shows maximum number of containerized applications that can be configured:

```
switch# show capacities containers
System Capacities: Filter CONTAINERS
Capacities Name                                     Value
-----
Maximum number of containerized applications configurable in the system  2
```

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A	Operator (>) or Manager (#)	
6400		
8100		
8320		
8325		
8325H		
8325P		
8360		
8400		
9300		
9300S		
10000		
10040		

show capacities-status containers

`show capacities-status containers`

Description

Reserved for future use.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A	Operator (>) or Manager (#)	
6400		
8100		
8320		
8325		
8325H		
8325P		
8360		
8400		
9300		
9300S		
10000		
10040		

show running-config container

`show running-config container`

Description

Shows the running configuration of all the containers.

Parameter	Description
container	Specifies that container running configuration must be displayed.

Examples

Shows the running configuration for the container:

```

container app1
  enable
  image-location http://30.0.0.2:8000/container.img vrf mgmt
  restrict cpu 10
  restrict memory 512
  network vrf mgmt
    port-map host-port 8000 container-port 9000 protocol tcp
  exit
  mount 1
    source usb /app1-data
    destination /app-data
  exit
  env PYP value /usr/bin/python3
  env encryptedVar1 encrypted ciphertext AQBapcmTCVdag
container app2
  enable
  image-location http://[2001::2]:8000/changeValidation_x86_t.img vrf mgmt allow-
unsigned
  restrict cpu 5
  restrict memory 256
  network vrf default
    port-map host-port 7819 container-port 7819 protocol udp
  exit
  mount 1
    source usb /app2-data1
    destination /app-data1
  exit
  mount 2
    source usb /app2-data2
    destination /app-data2
  exit
  env PYP value /usr/bin/python3
  env encryptedVar1 encrypted ciphertext AQBapcmTCVdag
  env encryptedVar2 encrypted ciphertext AQpY4V4v9UtDa==

```

Command History

Release	Modification
10.16.1000	The output of this command includes mount and network VRF details, if configured.
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8325H 8325P 8325P 8360 8400 9300 9300S 10000 10040	Operator (>) or Manager (#)	

vrf (container manager)

vrf <VRF-NAME>

Description

This command is deprecated. Use the [network vrf](#) command to define a VRF for a container.

Command History

Release	Modification
10.12	Command introduced

Command Information

Platforms	Command context	Authority
6300, except for S3L75A, S3L76A and S3L77A 6400 8100 8320 8325 8360 8400 9300 9300S 10000 10040	config-container-<CONTAINER-NAME>	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

HPE Aruba Networking Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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(docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.